

DIGITAL IMAGE PROCESSING

LECTURE # 2

APPLICATIONS OF IMAGE PROCESSING

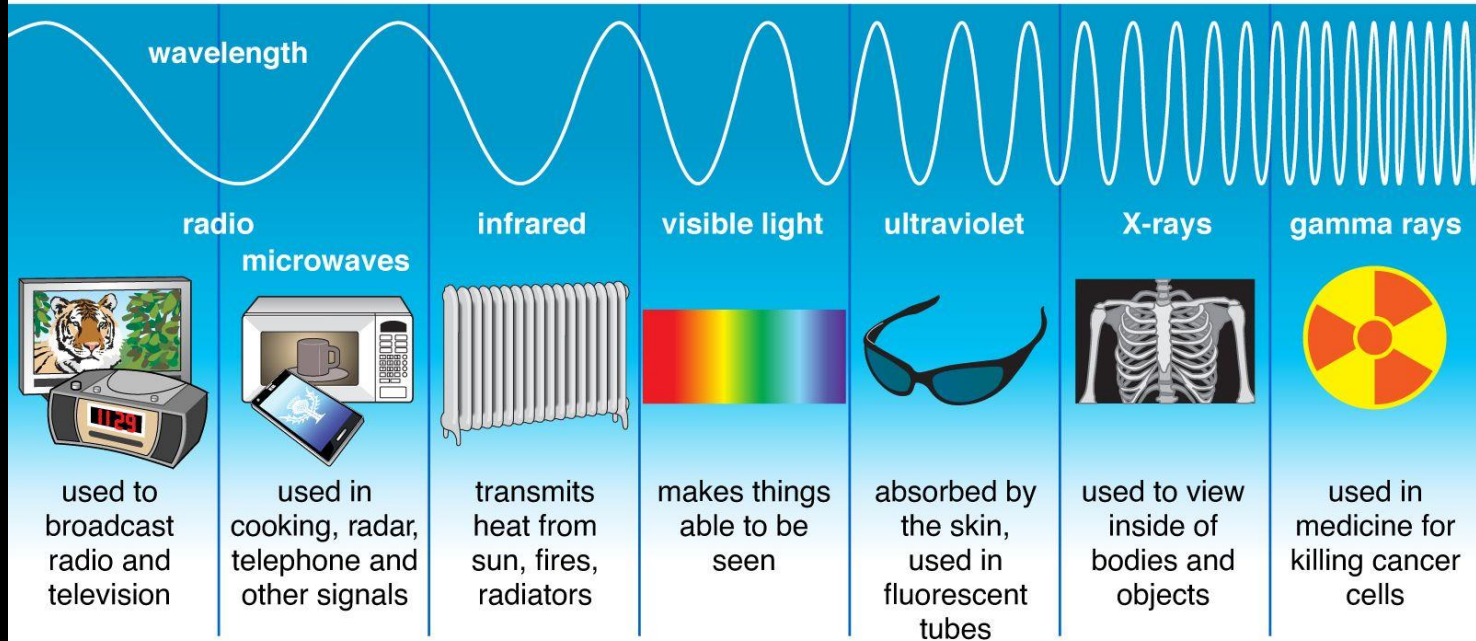
Contact Information

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EM Spectrum

Types of Electromagnetic Radiation

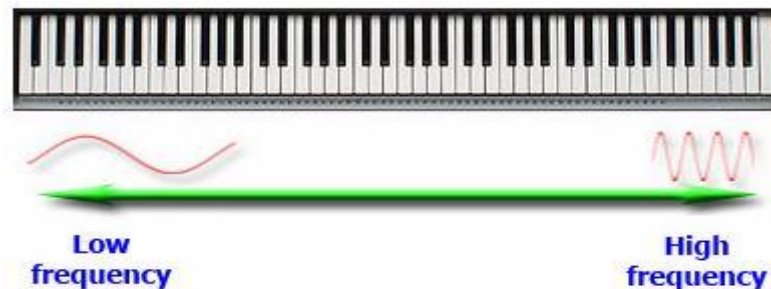


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EM Spectrum [1]

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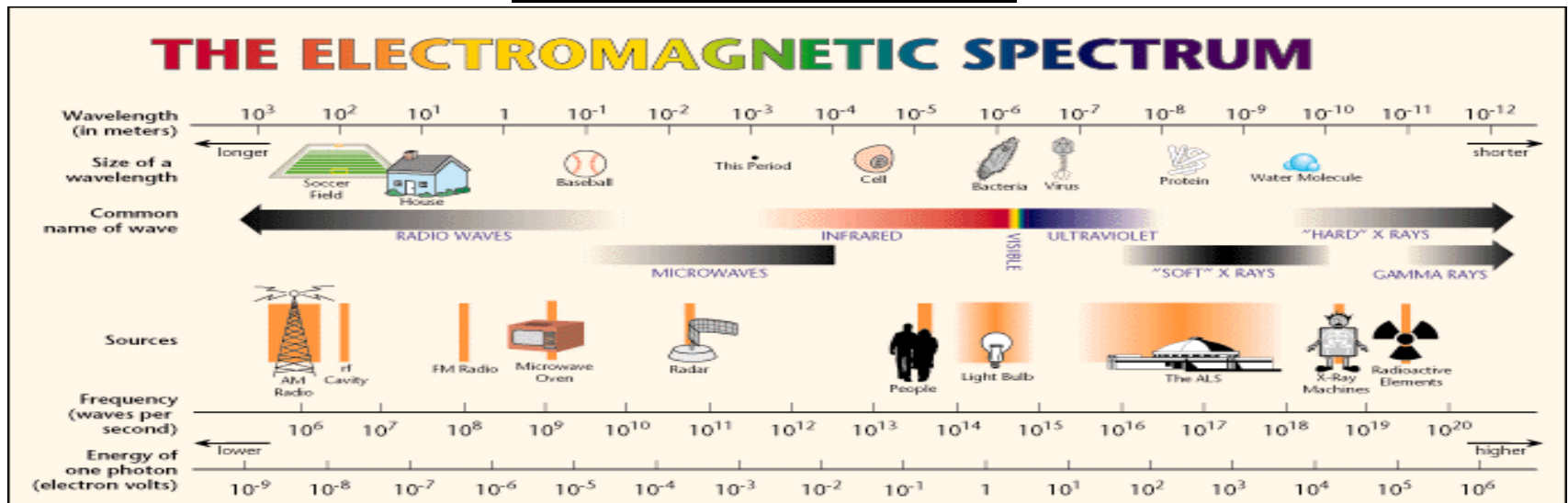
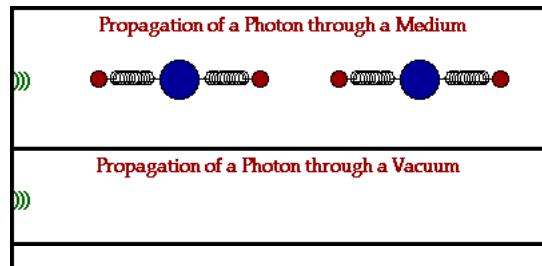
- ❑ Scientists have found that many types of waves can be arranged together like the notes on a piano keyboard, to form a scale.
- ❑ We can specify waves through frequency and wavelength
- ❑ The 'low notes' have a low frequency and a long wavelength.
- ❑ The 'high notes' have a high frequency and a short wavelength.
- ❑ With a sound wave, it's the air particles that are vibrating.
- ❑ So, what's vibrating when an electromagnetic wave passes by?
- ❑ Electromagnetic waves are vibrations of magnetic and electric fields. So, they don't need air to travel. They don't need anything to be there at all.



EM Spectrum

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- EM waves are mass less particles propagates at speed of light
- Human beings can see only the visible band portion
- Specific imaging systems have been designed for other bands of EM Spectrum



Gamma Rays [2]

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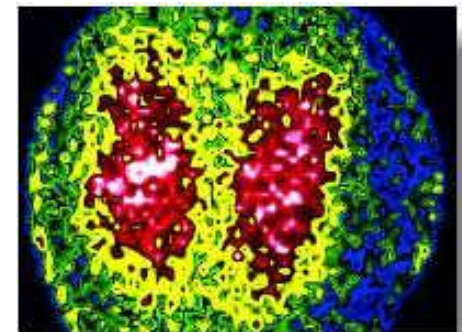
- ❑ Gamma rays are given off by **stars**, and by some **radioactive substances**.
- ❑ They are extremely high-frequency waves and carry a large amount of energy.
- ❑ They pass through most materials and are quite difficult to stop - you need lead or concrete to block them out.
- ❑ Because Gamma rays can kill living cells, they are used to **kill cancer cells** without having to conduct difficult surgery.
- ❑ This is called "**Radiotherapy**" and works because cancer cells can't repair themselves like healthy cells can when damaged by gamma rays. Getting the dose right is very important!

Gamma Rays [2]

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Tracers:

- ❑ Doctors can put slightly **radioactive substances into a patient's body**, then **scan the patient** to detect the gamma rays and build up a picture of what's going on inside the patient. This is very useful because they can see the body processes actually working, rather than just looking at still pictures.
- ❑ Example: the picture below is a "Scintigram" and shows an asthmatic person's lungs. The patient was given a slightly radioactive gas to breathe, and the picture was taken using a gamma imaging camera to detect the radiation.



X-Rays [3]

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- ❑ X-rays are very high frequency waves emitted by stars and some types of nebula, and carry a lot of energy. They will pass through most substances, and this makes them useful in medicine and industry to see inside things.
- ❑ An **X-ray machine** works by firing a beam of electrons at a "target". If we fire the electrons with enough energy, X-rays will be produced.
- ❑ X-rays are used by doctors to **see inside people**. They pass easily through soft tissues, but not so easily through bones.
- ❑ Lower energy X-Rays don't pass through tissues as easily, and can be used to scan soft areas such as the brain
- ❑ X-Rays are also used in **airport security** checks, to see inside your luggage.



Ultraviolet [4]

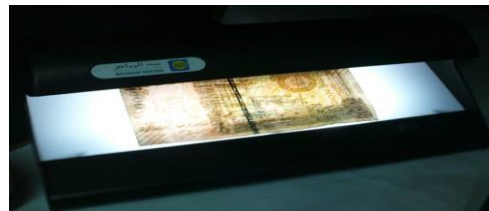
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- ❑ **Ultraviolet (UV)** light is electromagnetic radiation with a wavelength shorter than that of visible light, but longer than X-rays, in the range 10 nm to 400 nm.
- ❑ It is named because the spectrum consists of electromagnetic waves with frequencies higher than those that humans identify as the color violet.
- ❑ These frequencies are invisible to humans, but visible to several insects and birds.
- ❑ They are also indirectly visible, by causing fluorescent materials to glow with visible light.

Uses of Ultraviolet

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- ❑ Uses for UV light include getting a **suntan** [11], **detecting forged bank notes** [12], and hardening some types of dental filling.
- ❑ You also see UV lamps in clubs, where they make your clothes glow. This happens because substances in washing powder "fluoresce" when UV light strikes them - they absorb the UV and then re-radiate the energy at a longer wavelength.
- ❑ Hospitals use UV lamps to sterilize surgical equipment and the air in operating theatres. Food and drug companies also use UV lamps to **sterilize** their products [10].
- ❑ Suitable doses of Ultraviolet rays **cause the body to produce vitamin D**, and this is used by doctors to treat vitamin D deficiency and some skin disorders.
- ❑ Forensic scientists use UV light to reveal hidden blood and fingerprints.



Visible Light [5]

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- ❑ Our eyes can detect only a **tiny** part of the electromagnetic spectrum, called **visible light**.
- ❑ This means that there's a great deal happening around us that we're simply not aware of, unless we have instruments to detect it.
- ❑ Light waves are given off by **anything that's hot enough to glow**.
- ❑ This is how light bulbs work - an electric current heats the lamp filament to around 3,000 degrees, and it glows white-hot.
- ❑ The surface of the Sun is around 5,600 degrees, and it gives off a great deal of light.



Infrared [6]

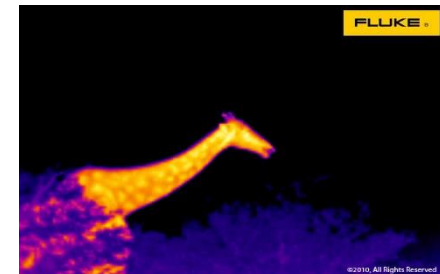
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- ❑ Infra red waves are just below visible red light in the electromagnetic spectrum ("Infra" means "below").
- ❑ You probably think of Infra-red waves as heat, because they're given off by **hot objects**, and you can feel them as warmth on your skin.
- ❑ Infra Red waves are also given off by **stars, lamps, flames** and anything else that's warm - including you.
- ❑ The detector on the security light picks up the Infra red radiation from your body.
- ❑ They are used for many tasks, for example, **remote controls for TVs** and video recorders, short-range communications between mobile phones.

Uses of Infrared

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- ❑ Because every object gives off IR waves, we can use them to "see in the dark". These devices are essential for wildlife research, security, and military operations.
- ❑ **Night sights** for weapons sometimes use a sensitive IR detector. Remember the movie, "Predator"?
- ❑ "Passive Infra-Red" **detectors** are used in burglar **alarm systems** that many people have fitted outside their houses for security. These detect the Infra-Red emitted by people and animals.
- ❑ You've probably seen TV programs in which **police helicopters** track criminals at night, using "**thermal imaging**" cameras that can see in the dark. These cameras use Infra-Red waves instead of "ordinary" light, which is why people look bright in these pictures. Similar cameras are also used by fire crews and other **rescue workers**, to find people trapped in rubble.



TeraHertz

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- ❑ Terahertz radiation is a region of the spectrum between far infrared and microwaves.
- ❑ Until recently, the range was rarely studied but applications such as imaging and communications are now appearing.
- ❑ Terahertz radiation can penetrate fabrics and plastics, so it can be used in surveillance, such as security screening, to uncover concealed weapons on a person, remotely.
- ❑ Terahertz radiation is strongly absorbed by atmospheric gases, making this frequency range useless for long distance communication.

Microwaves [7]

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- ❑ Microwaves are basically extremely high frequency radio waves and are made by various types of **transmitter**.
- ❑ In a mobile phone, they're made by a transmitter chip and an antenna, in a microwave oven they're made by a "magnetron".
- ❑ Stars also give off microwaves.
- ❑ Microwaves cause water and fat molecules to vibrate, which makes the substances hot. So, we can use microwaves to **cook** many types of food.



Uses of Microwaves

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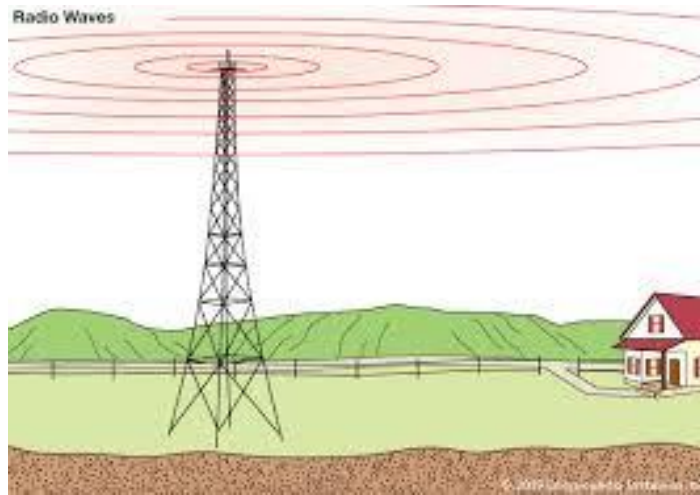
- ❑ **Mobile phones** use microwaves
- ❑ Microwaves are also used by fixed traffic **speed cameras**, and for **radar**, which is used by aircraft, ships and weather forecasters.
- ❑ The most common type of radar works by sending out bursts of microwaves, detecting the "echoes" coming back from the objects they hit, and using the time it takes for the echoes to come back to work out how far away the object is.



Radio Waves [8]

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- ❑ Radio waves are made by various types of **transmitter**, depending on the wavelength.
- ❑ They are also given off by stars, sparks and lightning, which is why you hear interference on your radio in a thunderstorm.
- ❑ Radio waves are the lowest frequencies in the electromagnetic spectrum and are used mainly for **communications**.
- ❑ MRI scans use radio waves to produce detailed images of organs.



Why do we Process Images?

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- ❑ Improvement of Pictorial information for human perception
- ❑ Image Processing for Autonomous Machine
- ❑ Facilitate picture storage and transmission
- ❑ Extract information from images
- ❑ Prepare for display or Printing

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Human Perception



Human perception

Understanding why we see what we do

Human Perception



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- ❑ Employ methods capable of enhancing pictorial information for human interpretation and analysis
- ❑ Typical Applications are::
 - ✓ Noise Filtering
 - ✓ Content Enhancement
 - Contrast Enhancement
 - Deblurring
 - ✓ Medical Imaging
 - ✓ Remote Sensing

Noise Filtering

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- Noise can be removed from the image by using the following filters
 - ✓ Linear Filters (Average Filters, Weighted average Filters)
 - ✓ Non- Linear Filters (Order Statistics Filters [Median, Minimum, Maximum])

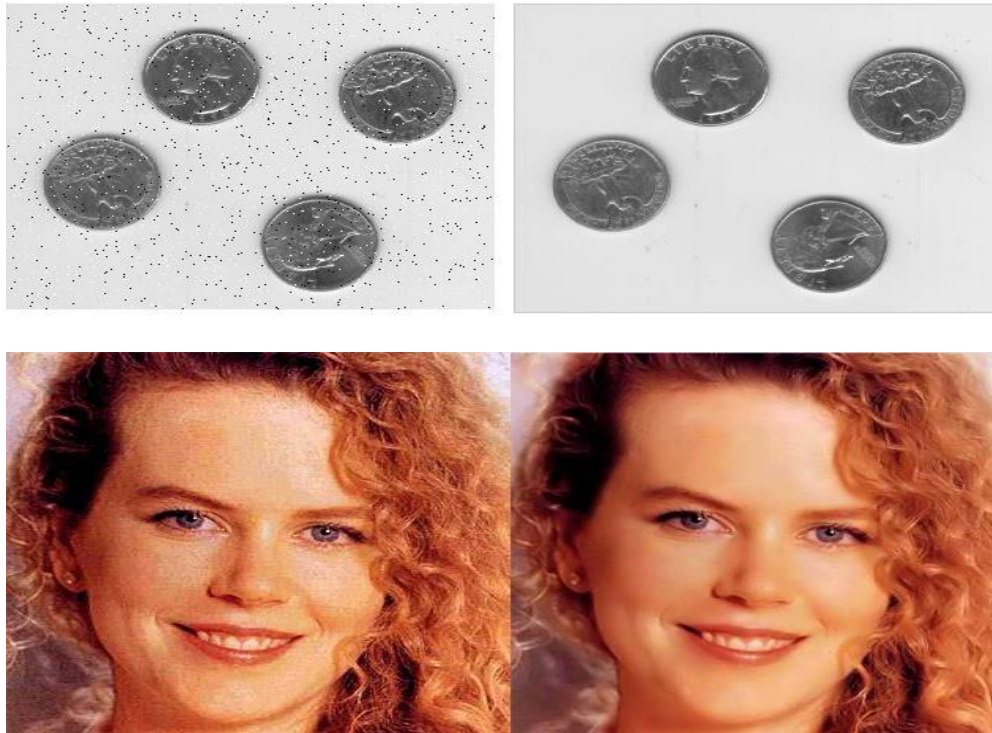


Image Deblurring

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- ❑ There are two primary kinds of blur in photos : focus blur and motion blur.
- ❑ Here are examples of focus and motion blur, respectively:
 - ✓ Focus blur happens when the subject of your photo is simply out of focus. The solution is to make sure your autofocus is on and try again.
 - ✓ Motion blur, on the other hand, doesn't happen because your subject is out of focus. It happens because your subject is moving relative to the camera frame while the exposure is being made



Focus Blurred Image

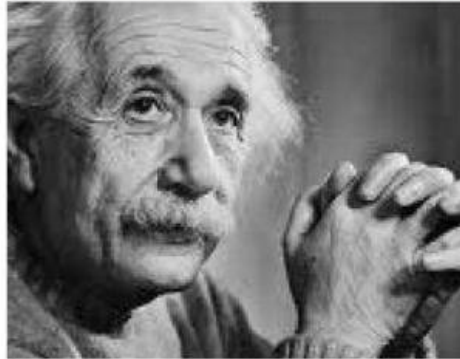
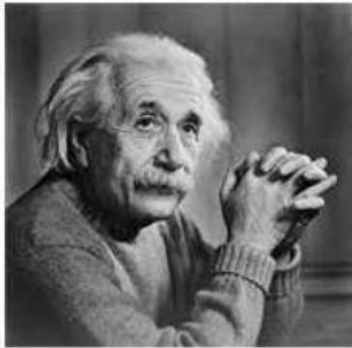


Motion Blurred Image

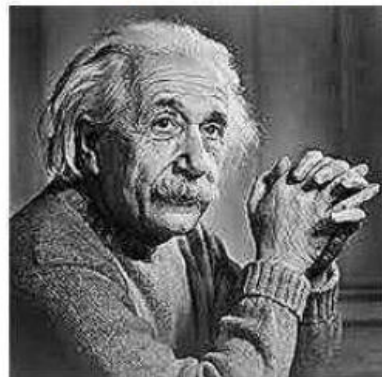
Image Sharpening

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The original image The zoomed image Blurr image



Sharp image



Edges



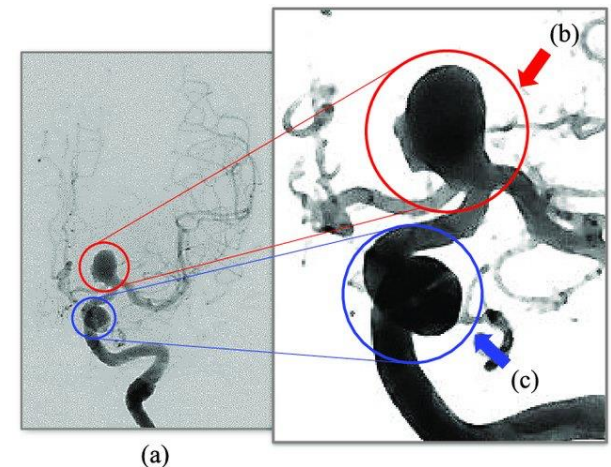
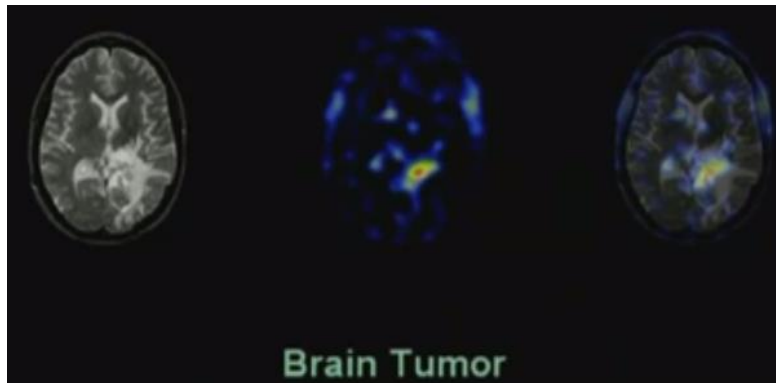
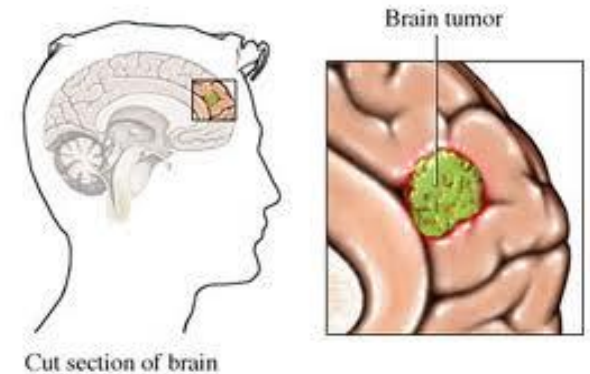
Medical Imaging

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□ Image processing has been widely used in the field of medical

□ Applications of Image processing in medical are:

- ✓ Tumor Detection
- ✓ Brain Aneurysm Detection
- ✓ Cancer Detection
- ✓ Ultrasound and many more.....
- ✓ Mirai and AsymMirai are AI tools capable of predicting cancers 5 years before [20]



Remote Sensing

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- ❑ Remote sensing can be defined as any process whereby information is gathered about an object, area or phenomenon without being in contact with it.
- ❑ Our eyes are an excellent example of a remote sensing device. We can gather information about our surroundings by gauging the amount of the visible light reflected from some external source as it reflects off objects in our field of view.
- ❑ Contrast this with a thermometer, which must be in contact with the phenomenon it measures, and thus is not a remote sensing device.

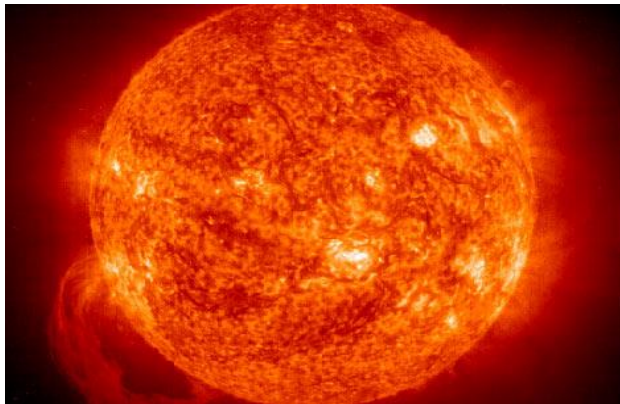


Remote Sensing

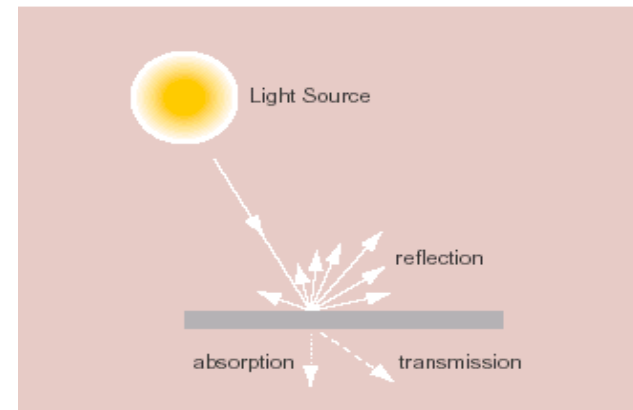
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□ Fundamental Considerations

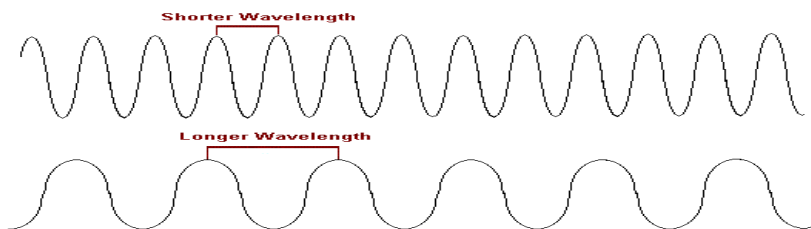
Energy Source



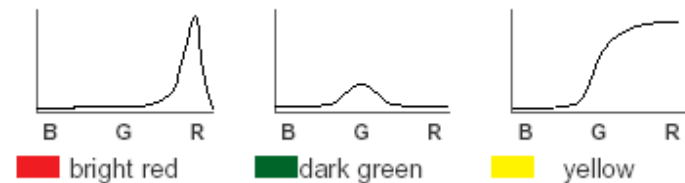
Interaction Mechanisms



Wavelength



Spectral Response Patterns

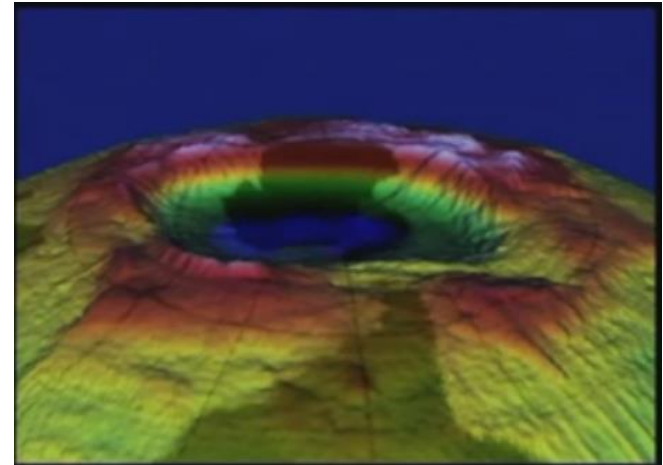
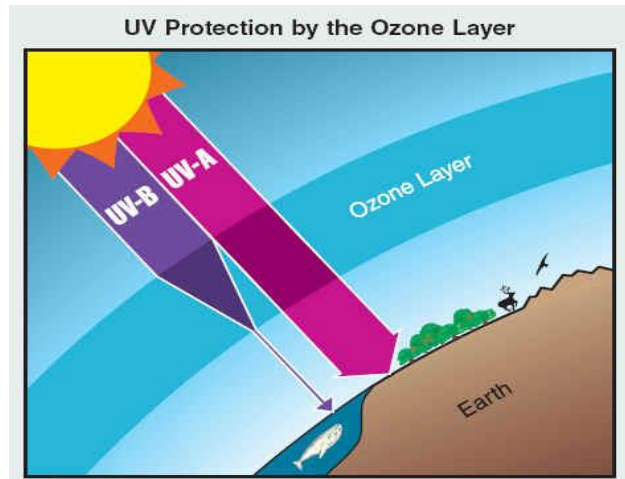


Atmospheric Study

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Image Processing can also be used for atmospheric study to analyze:

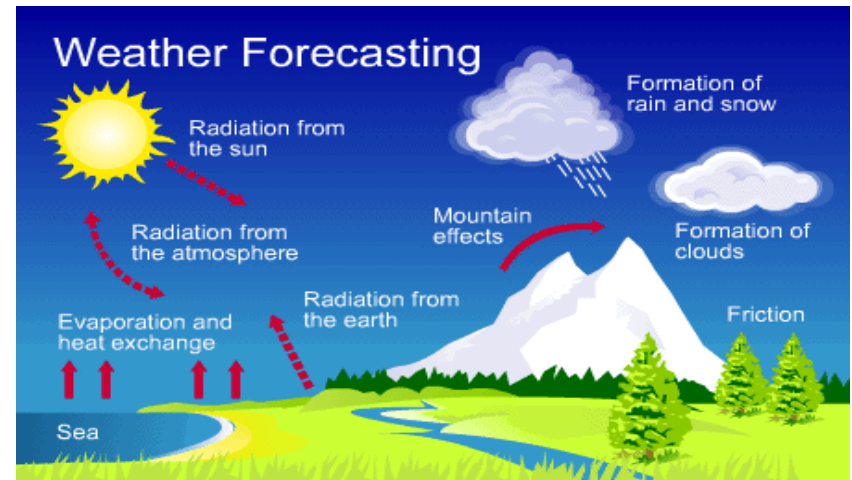
- ✓ Atmospheric Visibility Monitoring
- ✓ Ozone Layer Study
- ✓ Climatology e.t.c



Weather Forecasting

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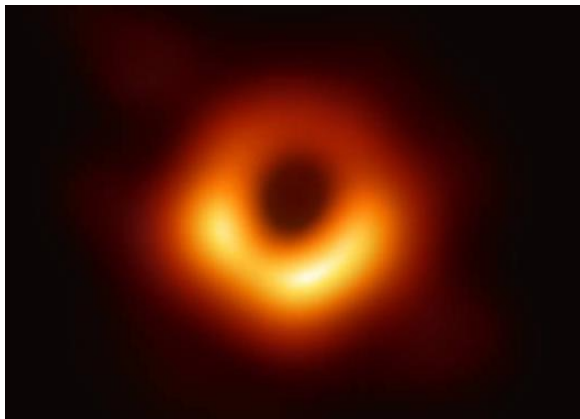
- ❑ Image processing techniques have been used extensively for weather forecasting
- ❑ Techniques like Image Enhancement and Restoration are used significantly to obtain better quality images free from noise or any degradation
- ❑ Techniques like Image Segmentation uses to partition specific parts like the clouds from the image
- ❑ Image Recognition has been applied to classify various weather phenomena's e.g to classify the clouds from the hurricane



Astronomy

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- ❑ Image processing techniques have been used extensively for Astronomical Observations
- ❑ Astronomy is a natural science that deals with the study of celestial objects (such as stars, planets, comets, nebulae, star clusters, and galaxies) and phenomena that originate outside the Earth's atmosphere
- ❑ Blackhole Image Capture [19]



Machine Vision Applications

JADAK Vision
Software



Embedded / OEM
Smart Cameras



Fixed Mount
Smart Cameras



JADAK Vision
Services & Solutions



Machine Vision

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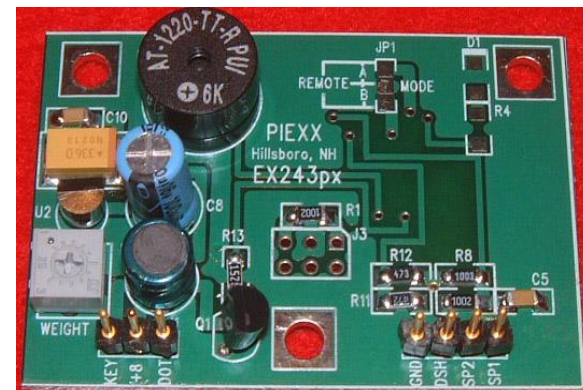
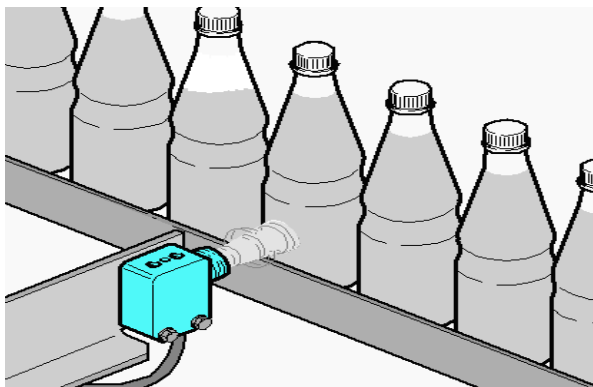
- ❑ Here the interest is on the procedures for extraction of image information suitable for computer processing
- ❑ Typical applications are:
 - ✓ Industrial Machine Vision for Product Assembly and Inspection
 - ✓ Automated Target Detection and Tracking
 - ✓ Finger Print Recognition
 - ✓ Iris Recognition
 - ✓ Face Detection, Tracking and Recognition e.t.c



Automated Product Inspection

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- ❑ Image Processing techniques have great utilization in the industry for various product inspections
- ❑ Some of them can be:
 - ✓ Automated Inspection of Bottling Plant Automation
 - ✓ Automated Inspection of IC Manufacturing
 - ✓ Automated Inspection of Computer Components e.t.c

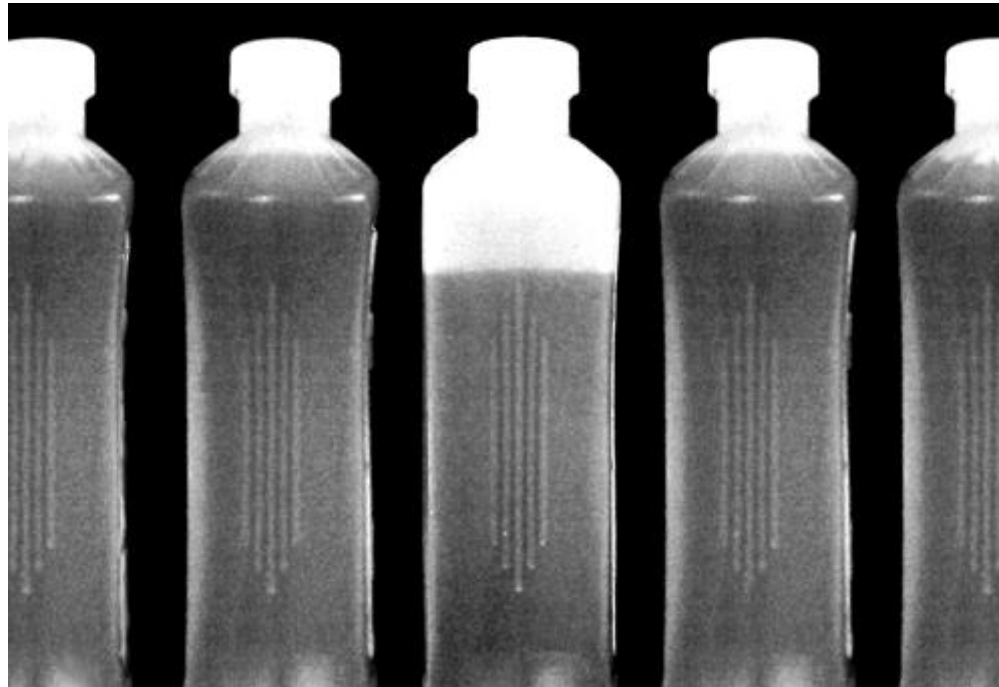


Automated Inspection of Bottling Plant

Automation

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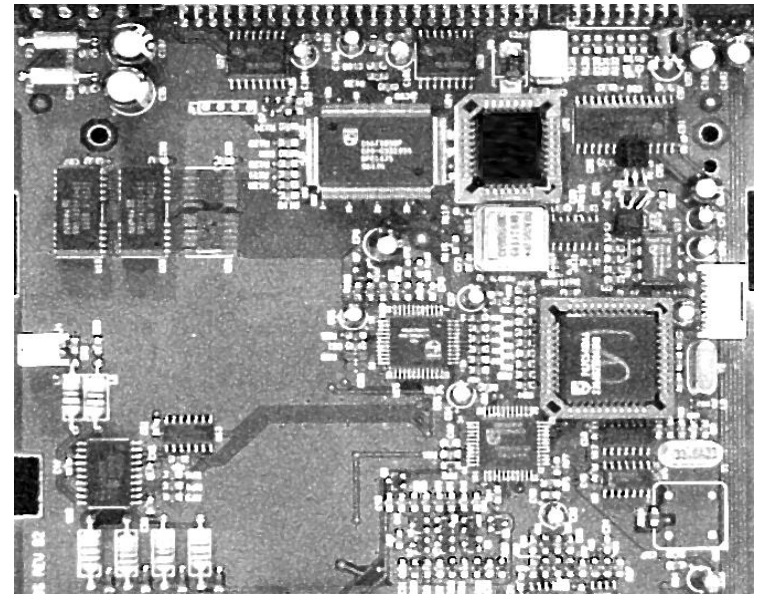
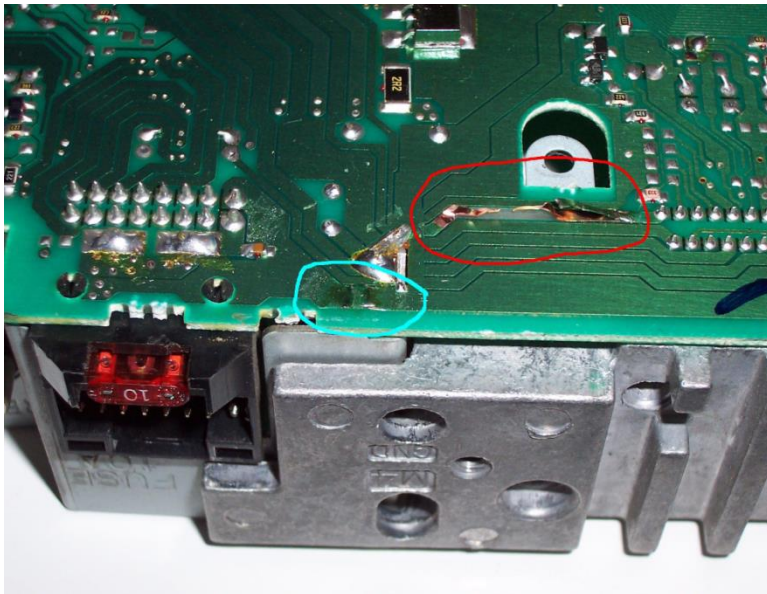
- ❑ Image processing techniques can be used to inspect the bottles of soft drinks to check whether any bottle is empty or partially filled so to avoid any such product delivered to the customer which can affect the goodwill of the company



Automated Inspection of Integrated Circuits & Circuit board/ Motherboard

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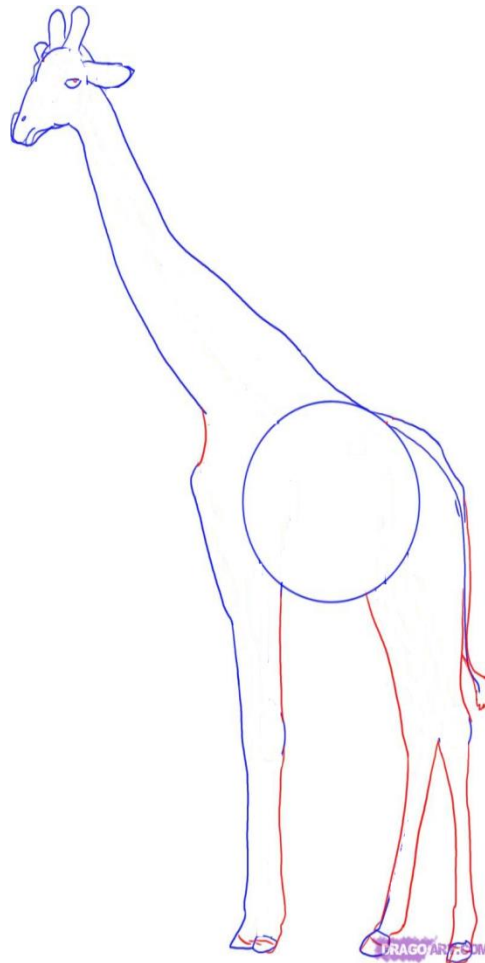
- Image processing techniques can be used to inspect the Integrated Circuits during the manufacturing phase to detect any missing components or any parts that is broken
- e.g In 1st figure some part is broken while in second figure a component is missing so these problems can be identified through image subtraction method



Boundary Information

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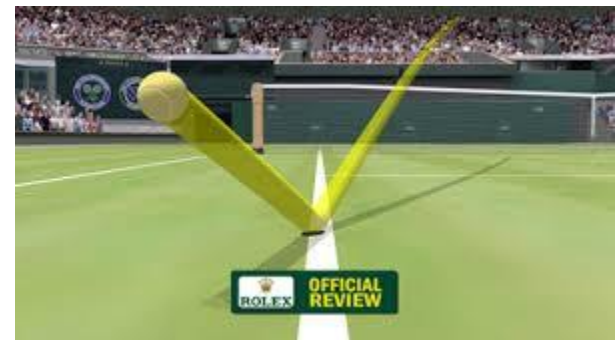
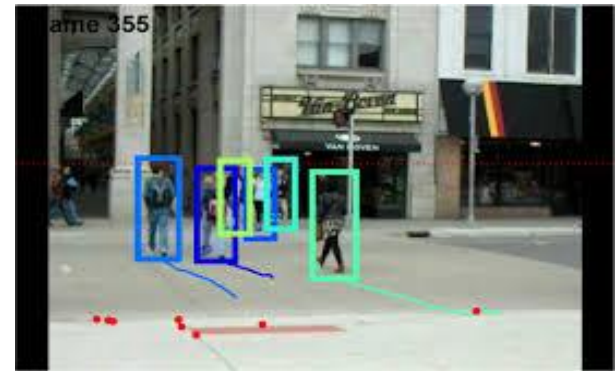
- ❑ The boundary of an object is very useful to recognize the object



Video Sequence Processing

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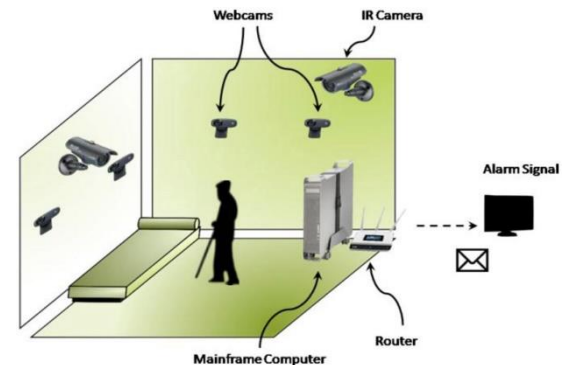
- ❑ The major emphasis of image sequence processing is detection of moving parts
- ❑ Typical applications are:
 - ✓ Detection and tracking of moving targets for security surveillance purpose
 - ✓ To find out the trajectory of a moving target
 - ✓ Monitoring the movements of organ boundaries in medical applications



Movement Detection and Tracking

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- ❑ There are huge number of applications in the field of corporate sector, educational institutions, sports industry, etc., which involves the detection and tracking of an object to achieve the task
- ❑ For example, some of the applications can be::
 - ✓ The detection and tracking of pedestrians to give information to drivers who drives vehicle on the road to avoid as many accidents as possible
 - ✓ The detection and tracking of suspicious person to achieve the security measures
 - ✓ The detection and tracking of old persons or babies in the houses to monitor their activities and includes an alarming system to inform the other house members in case of any dangerous situation
 - ✓ The detection and movement of instructor in front of the white board in case you are developing a system to extract the information from the white board into handouts from the video lectures etc.



Fingerprint Recognition [13, 14]

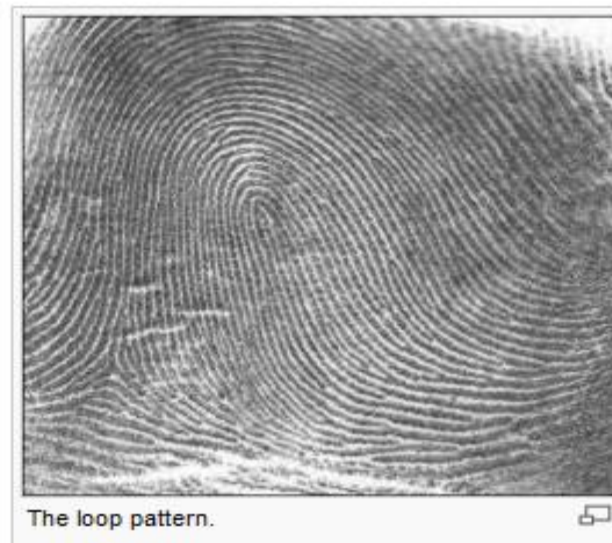
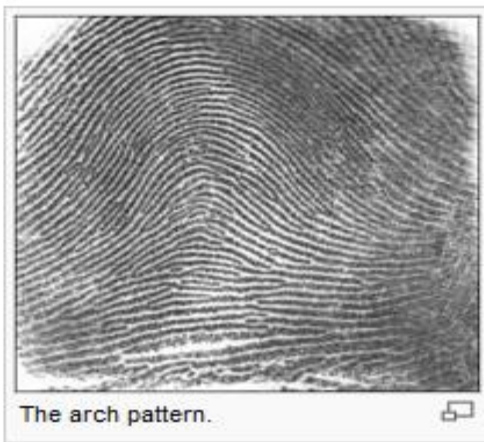


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Patterns

The three basic patterns of fingerprint ridges are the arch, loop, and whorl:

- ✓ arch: The ridges enter from one side of the finger, rise in the center forming an arc, and then exit the other side of the finger.
- ✓ loop: The ridges enter from one side of a finger, form a curve, and then exit on that same side.
- ✓ whorl: Ridges form circularly around a central point on the finger.



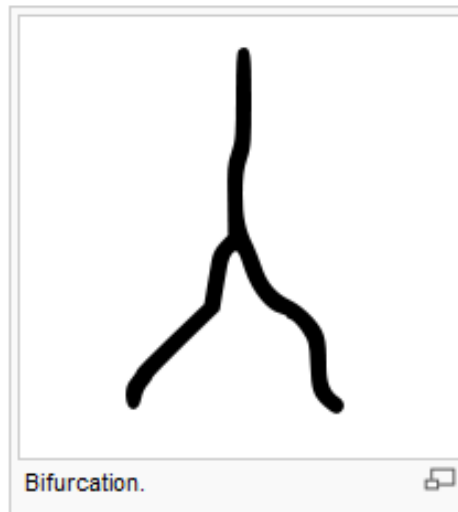
Fingerprint Recognition [13, 14]

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Minutia

The major Minutia features of fingerprint ridges are ridge ending, bifurcation, and short ridge (or dot).

- ✓ The ridge ending is the point at which a ridge terminates.
- ✓ Bifurcations are points at which a single ridge splits into two ridges.
- ✓ Short ridges (or dots) are ridges which are significantly shorter than the average ridge length on the fingerprint.



Face Detection

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❑ Viola and Jones Face Detection Model [17]

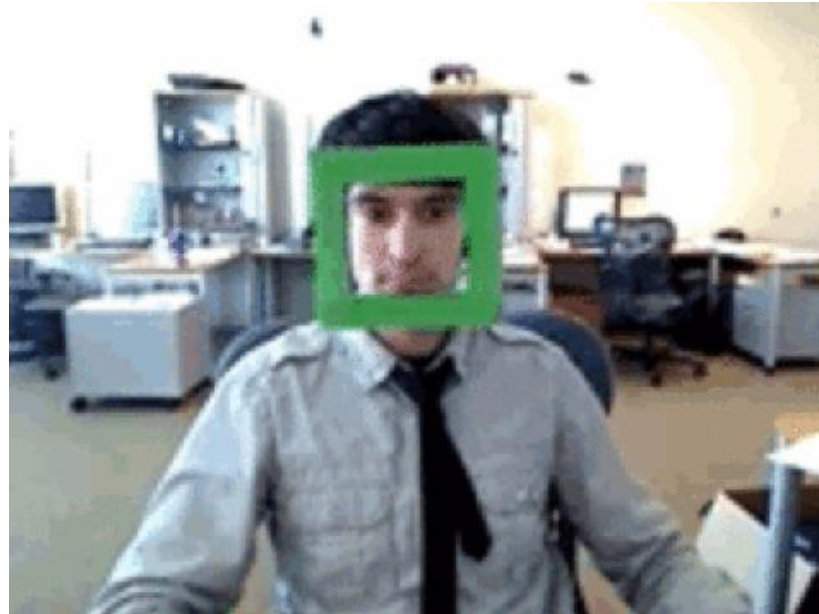
- ✓ Haar Features
- ✓ Integral Image
- ✓ Adaboost
- ✓ Cascading

❑ Face Detectors [21]



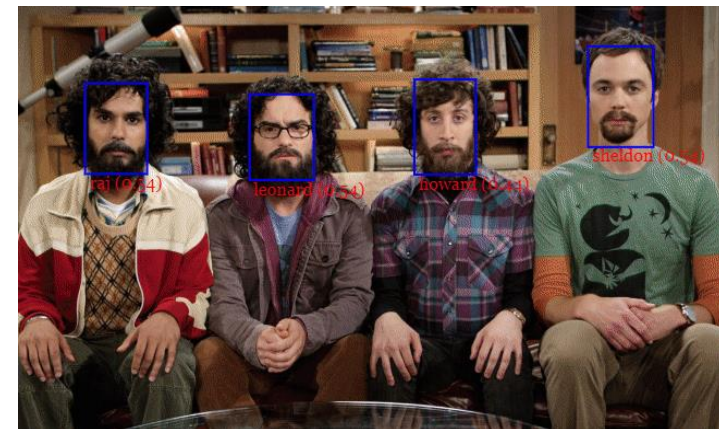
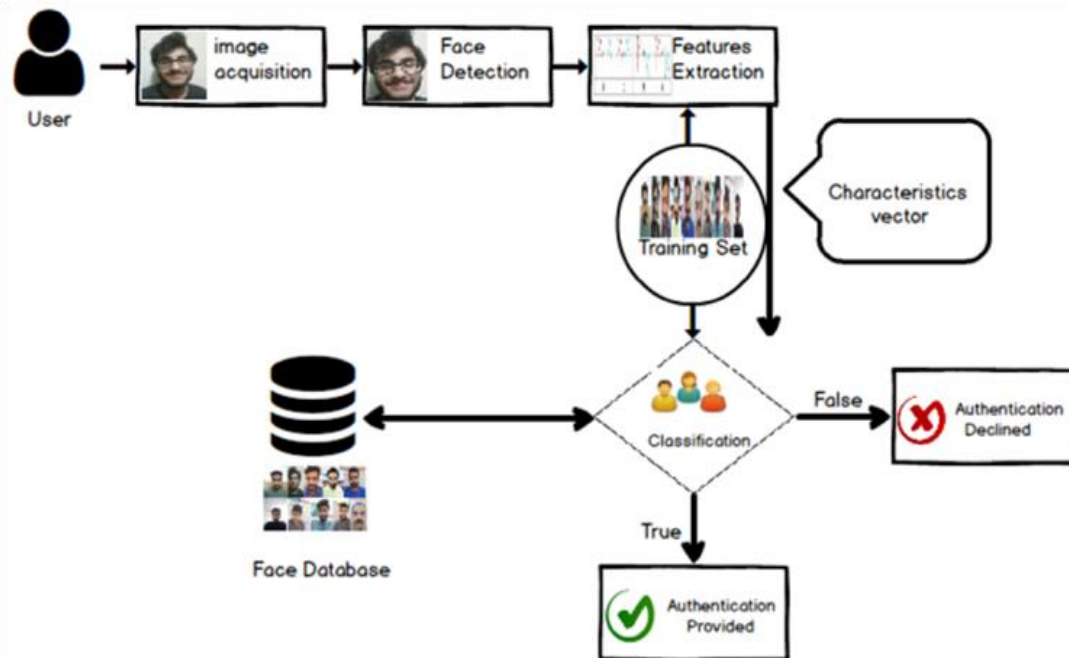
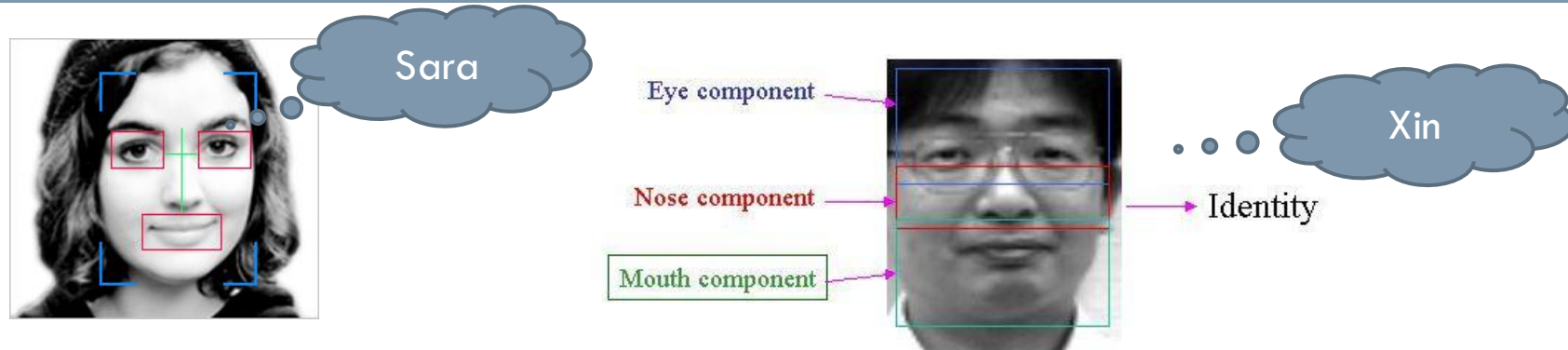
Face Tracking

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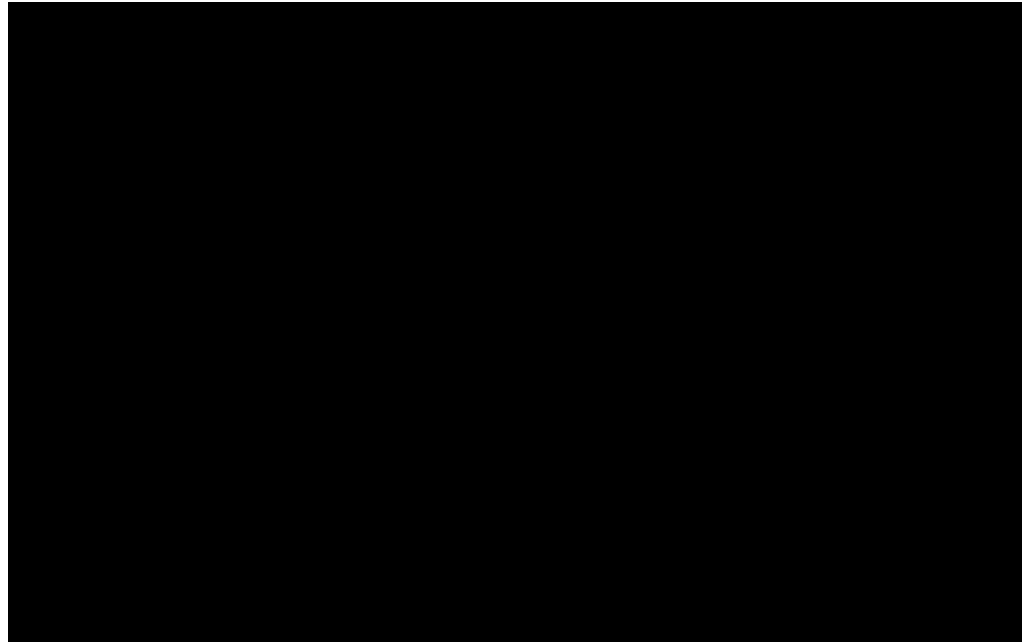
Face Recognition

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Face Detection, Tracking and Recognition

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Personal Identification Using Iris Recognition

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HOW IRIS SCANNERS RECORD IDENTITIES

- 1 Scanner reads from outer iris inwards to pupil edge
 - 2 Scanner plots distinct markings on iris and maps unique shape
 - 3 After plotting many marks within the iris all data is saved to a database
 - 4 Other scanners will compare this data to verify individual identities
-
- The diagram shows a detailed illustration of a human eye. Overlaid on the iris is a grid of concentric circles and radial lines, with small red crosses marking specific points. This visualizes the process of scanning the iris by plotting distinct markings.

Your Body is your password

Deepfakes [15]

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❑ Face-swap: in which the face in a video is automatically replaced with another person's face



❑ Lip-Sync: in which a source video is modified so that the mouth region is consistent with an arbitrary audio recording.



❑ Puppet-Master: in which a target person is animated (head movements, eye movements, facial expressions) by a performer sitting in front of a camera and acting out what they want their puppet to say and do.



Deepfakes [15]

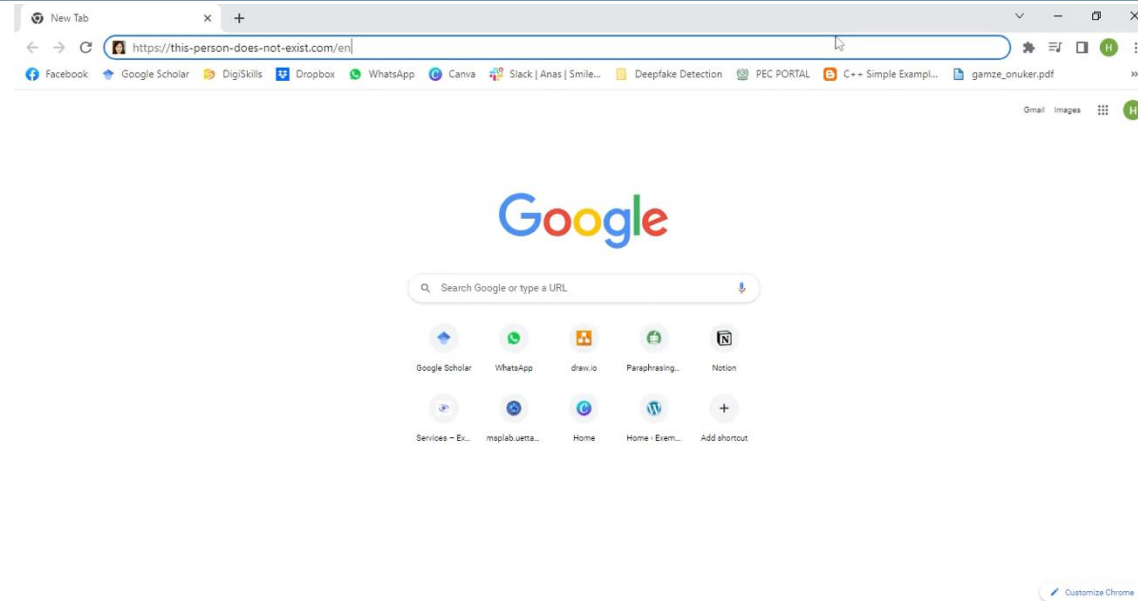
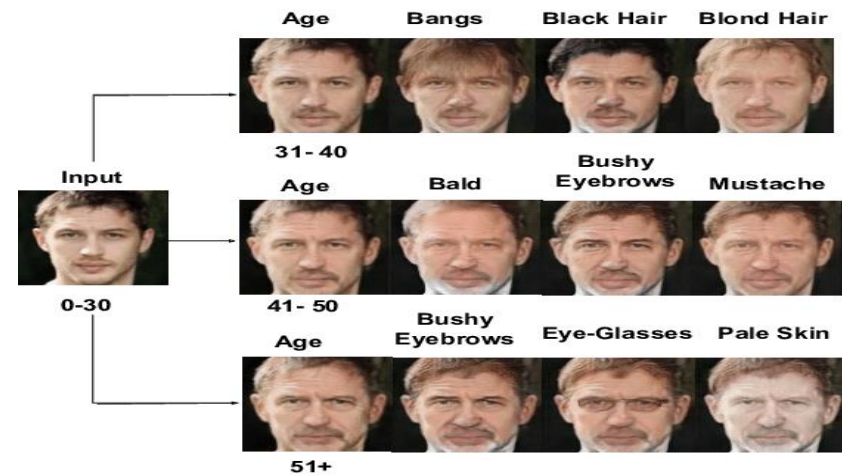
46

❑ Entire Face Synthesis [18]



A few sample faces — all completely fake — created by ThisPersonDoesNotExist.com

❑ Facial Attribute Manipulation



References

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1. <http://www.darvill.clara.net/emag/index.htm>
 2. <http://www.darvill.clara.net/emag/emaggamma.htm>
 3. <http://www.darvill.clara.net/emag/emagxray.htm>
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